Linear Algebra (MT-1004)

Homework#16

Instructor: Muhammad Ali

Question 1

1. Find the projection of $\mathbf{u} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ onto $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.
2. Determine the component of $\mathbf{a} = \begin{bmatrix} 4 \\ -2 \end{bmatrix}$ orthogonal to $\mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.
3. Prove that $\text{proj}_{\mathbf{v}}(\mathbf{u})$ is parallel to $\mathbf{v}$.
4. **Python Task:** Write a Python function to compute the projection of one vector onto another.

```
def projection(u, v):
    # Your code here
```